

JAMES BERNARD AMOAH FEI-BAFFOE

[James Website](#) • jbfeib27@colby.edu • [LinkedIn](#) • [Github](#) • (207)-830-1361

EDUCATION

Colby College - Waterville, Maine, USA

Graduating May 2027

Majors: Computer Science and Mathematics Major GPA: 4.0

Relevant Coursework: Computational Thinking: Python, Data Structures, Algorithms and Object-Oriented Programming: Java, Calculus I, Calculus II, LinAlg, Data Analysis and Visualization, Software Engineering, Operating Systems, Programming Languages, Analysis of Algorithms, Neural Networks, Parallel Computing (MIT)

Awards/Membership: National Science and Mathematics Quiz SemiFinalist, Dean's List, ColorStack Member
1st Place - Code for Good Hackathon, JPMorgan Chase (Oct 2025), Two Sigma Software Engineering Mentee

EXPERIENCE

SWE Intern @ LinkedIn

May 2026 - August 2026 Sunnyvale, CA

- Shipped LLM agent infrastructure (tool-calling, multi-step planning, RAG over 10+ HR data sources) powering an AI assistant for 20K+ employees, automating onboarding, transfers, and benefits workflows end-to-end
- Engineered low-latency inference pipeline on LinkedIn's distributed stack (Kafka, Pinot, gRPC), driving p99 below 800ms via prompt caching, speculative tool execution, and async fan-out
- Built eval harness and guardrails (hallucination, PII leakage), enabling safe rollout to HR operations

SWE Intern @ Gusto

May 2025 - September 2025 San Francisco, CA

- Built automated payment recovery system processing 100K+ failed ACH transactions, saving \$1.25M annually in manual processing costs
- Designed SQL queries to detect orphaned bank accounts within regulatory deadlines while maintaining transaction safety
- Implemented production monitoring with DataDog, ensuring zero disruption to critical banking operations

SureStart, Machine Learning Engineer Intern

May 2024 - August 2024 Waterville, ME

- Built sentiment analysis pipeline using BERT and PyTorch to process 50K+ customer service tickets
- Developed CNN image recognition system using TensorFlow, achieving 0.89 F1-score on 10K labeled images

Lead Developer & First Place Winner, Code for Good Hackathon JPMorganChase

October 2025 Plano, TX

- Led development of full-stack donation platform using Next.js, React, and PostgreSQL with real-time messaging features
- Integrated Stripe payment processing with webhook verification and idempotent transaction handling for donations up to \$100K
- Developed backend APIs and integrated AI-powered newsletter generation for donor outreach

Colby College CS152 Lab Assistant

February 2024 - May 2024 Waterville, ME

- Conducted code reviews and debugging sessions for 60+ students in a Data Structures and Algorithms course

PROJECTS

Horizon: Financial SaaS Platform

- Developed Horizon, a comprehensive financial SaaS platform using Next.js and TypeScript, incorporating Plaid and Dwolla for real-time transaction processing and multi-bank connectivity.
- Implemented server-side rendering authentication with Clerk and input validation using React Hook Form with Zod schemas for type-safe form handling
- Engineered a responsive user interface with TailwindCSS and Chart.js, featuring visualizations of financial metrics and a real-time update system for seamless user experience across devices.

CPU Performance Monitoring System

- Engineered distributed Flask-based monitoring system tracking CPU/memory metrics across 20+ containerized microservices using Docker and Kubernetes on AWS EKS, implementing custom Prometheus exporters for real-time alerting
- Optimized deployment pipeline reducing container startup time from 45s to 8s through multi-stage Docker builds and ECR caching strategies, enabling faster incident response

SKILLS

- Programming languages:** Python, Java, Javascript, TypeScript, Solidity, Ruby, Ruby on Rails, C, C++
- Worked with:** Flask, Next.js, React, TailwindCSS, React Hook Form, Docker, Kubernetes, AWS EKS, AWS ECR, TensorFlow, PyTorch, Chart.js, SQL, Plotly.js, Plaid, PostgreSQL and Dwolla.